03 The Fate Room

The fate of a single new mutation

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 31 min	**+ 30 min**	The Copying Room (01) helps but is not needed.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/evolutionary-laboratories/#fate`

A shape is copied once with an error. That copy is a new mutation, carried by one individual among N. Two populations then run it forward until it is lost or has replaced every other copy. This guide is about how rarely the second ending happens.

LEARNING OBJECTIVES

After the core path you should be able to

- Explain that every new mutation starts as a single copy, at frequency $1/N$.
- Predict that a new neutral mutation is usually lost, and quickly.
- Use the tally to estimate the probability of survival, and relate it to $1/N$.
- Explain why a lost mutation is not evidence that it was harmful.

The extension adds

- Use $u = 1/N$, $\mathrm{E}[T_{\mathrm{fix}}] \approx 2N$ and $\mathrm{E}[T_{\mathrm{loss}}] \approx 2 \ln N$ to predict the room's numbers.
- Explain why the tally is so variable from one press to the next.
- Predict how much a selective advantage would change the survival chance, and test it in the Selection Room.

SUGGESTED ROUTES

- **35-minute practical (BSc):** steps 1–4 and questions 1–3.
- **Lecture demonstration (10 minutes):** step 2, asking the class to guess the tally before pressing.
- **MSc:** steps 5–7 and questions 4–6; step 7 leads straight into the Selection Room.

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 A mutation appears

5 min

SET UP
Step 1 of the room. Any starting shape.

PREDICT FIRST
The shape will be copied once with one random error. Will you be able to see the difference easily?

RUN
Press **Mutate** a few times, looking at the third panel where the parent sits faded behind its copy.

EXPLAIN
In what sense is the copy a *mutant*? Is the change good, bad, or neither?

2 One in twenty

10 min

SET UP
Step 2 of the room. Population size N = **20**.

PREDICT FIRST
The mutant is one individual among 20, and is neutral. What is the chance it ends up carried by everyone? How many populations do you think the room will need to run before it finds one where the mutation survives?

RUN
Press **Simulate evolution through time**. Read the tally beneath the two populations. Do it three times and write down each tally.

LOOK AT
the tally, the two populations, and the allele-frequency graph.

EXPLAIN
Compare your three tallies with your prediction. Why do they differ from one press to the next?

3 One in a hundred

10 min

SET UP
N = **100**.

PREDICT FIRST
How will the tally change? Write a number.

RUN
Simulate three times and record the tallies.

EXPLAIN
How does the tally scale with N? State the rule in one sentence.

4 Quick losses, slow wins

6 min

SET UP
Any N; 50 is a good choice.

PREDICT FIRST
Which population will finish first: the one where the mutation is lost, or the one where it takes over?

RUN
Simulate and watch the generation counters of the two populations.

EXPLAIN
Why is the losing population over almost at once, while the winner grinds on? What does this say about how much new variation is present in a population at any moment?

EXTENSION · MSC

5 Check the theory

12 min

SET UP
N = 12, 20, 50 and 100 in turn.

PREDICT FIRST
Fill in a table before you run: for each N, the survival probability $1/N$, the expected time to fixation for a winner ($\approx 2N$ generations) and the expected time to loss for a loser ($\approx 2 \ln N$).

RUN
Simulate twice at each N, recording the tally and the two finishing times.

EXPLAIN
Which of the three predictions is matched most closely by single runs, and which least? Why?

6 Why the tally jumps around

10 min

SET UP
N = 20.

PREDICT FIRST
If each population has a 1-in-20 chance of a win, the tally follows a geometric distribution with mean 20. Roughly how spread out will five tallies be?

RUN
Simulate five times and record the tallies. Compute their mean and range.

EXPLAIN
The standard deviation of a geometric distribution is nearly as large as its mean. How should that change the way you read any single tally?

7 Give it an advantage

8 min

SET UP

Go to the **Selection Room**: Haploid, $N = 100$, initial $p = 0.01$ (one copy), $s = 0.10$.

PREDICT FIRST

Neutral, the mutation would survive one time in a hundred. With a 10% advantage, how many of 10 runs will fix it?

RUN

Press **Run 10 Simulations** there, twice, and count the runs that fix A1.

EXPLAIN

Compare with the approximation $u \approx 2s$. Why is a clearly beneficial mutation still lost most of the time?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc Most new mutations are lost. Does that mean most mutations are harmful? Explain using this room.
2. BSc Every allele in every population began as a single copy in a single individual. Why does that make the allele frequencies we see today so remarkable?
3. BSc A conservation population of 15 individuals carries a rare neutral variant. Is it likely to keep it? Why?
4. MSc Show that the expected number of new neutral mutations that eventually fix per generation is independent of N (the rate of neutral substitution equals the mutation rate). Use the room's numbers.
5. MSc Why is the time a winner takes ($\approx 2N$) so much longer than the time a loser takes ($\approx 2 \ln N$)?
6. MSc What changes if the population is diploid rather than haploid?

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** The copy differs in one small, random way. It is neither good nor bad: there is no environment here to judge it.
- **Step 2.** Survival probability is 1/20. The tally is often near 20 but varies widely — anything from a handful to fifty or more is normal.
- **Step 3.** At $N = 100$ the tally is typically around 100. The rule: the number of populations needed is about N, because the chance of survival is $1/N$.
- **Step 4.** The losing population usually finishes within a few generations; the winner takes many tens. Most of the new variation in a population at any time is about to vanish.
- **Step 5.** At $N = 20$: $u = 0.05$, $T_{\text{fix}} \approx 40$, $T_{\text{loss}} \approx 6$. At $N = 100$: 0.01, about 200 and about 9. Single runs scatter widely around all three, least for the loss time.
- **Step 6.** Five tallies at $N = 20$ typically range from under 10 to over 40, with a mean near 20.
- **Step 7.** With $s = 0.1$, $u \approx 1 - e^{-0.2} \approx 0.18$: about 2 of 10 runs fix it, against 0.1 expected without the advantage. The advantage is real but drift still decides most outcomes while the mutation is rare.